

Transformer-Based Sleep Stage Classification and Personalized Sleep Health Profiling

Using Fast SCFormer-U, Evidential Learning and Soft-Viterbi Sequence Decoding



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Introduction and Motivation

The engineering problem

Score EEG into Wake, N1, N2, N3 and REM, then turn the sequence into an interpretable sleep profile.

Why uncertainty and sequence context?

Epoch predictions can fluctuate. Evidence exposes ambiguity; training-derived transitions support coherent decoding.

Why personalized profiling?

Recording-level indicators summarize sleep patterns beyond a single accuracy score; their research validity must be checked against labels.

Academic research system; sleep profiles are not a medical diagnosis.

Project Objectives

1 Five-stage classification

Build a compact four-channel CNN–self-attention–BiLSTM classifier.

2 Uncertainty and temporal refinement

Use evidential probabilities and uncertainty; decode each recording independently with training-only transitions.

3 Verified research profiles

Compute source-defined sleep indicators and compare predicted profiles with ground truth.

Literature Study

Study	Dataset	Modality	Protocol	Accuracy / macro F1
SeqSleepNet [1] 2019	MASS; 200	EEG+EOG+EMG	20-fold subject CV	87.10% / 83.30%
DeepSleepNet [2] 2017	Sleep-EDF; 20	EEG Fpz–Cz	20-fold subject CV	82.00% / 76.90%
Current raw model 2026	HMC subset; 24	Four-channel EEG	Fixed 16/4/4 split	53.75% / 55.12%

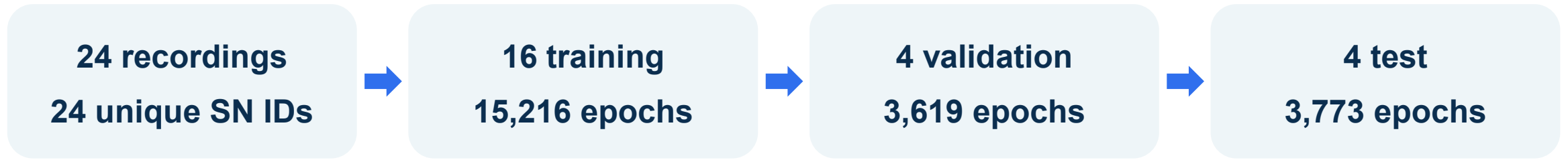
Context only: different datasets, modalities and split protocols prevent direct ranking.

[1] DOI: 10.1109/TNSRE.2019.2896659

[2] DOI: 10.1109/TNSRE.2017.2721116

HMC dataset: DOI 10.13026/t79q-fr32. Full sources and protocols in references.txt.

Dataset and Preprocessing



22,608 usable epochs • 30 s/epoch • 256→100 Hz

50 Hz notch • 0.3–35 Hz bandpass • per-channel epoch standardization

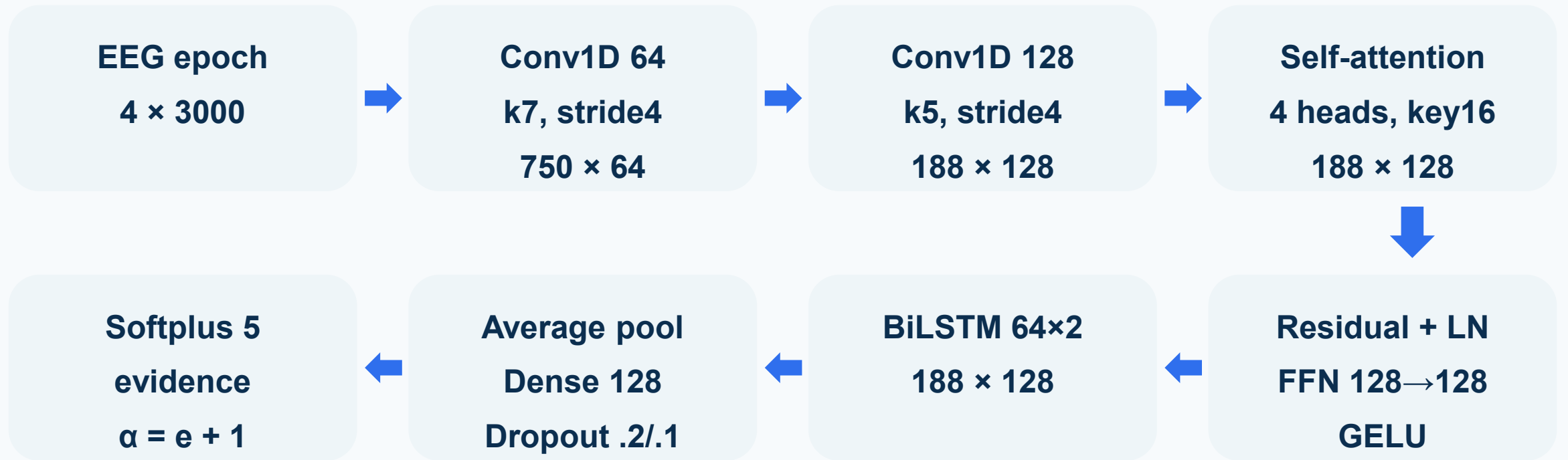
Selected EEG channels: F4–M1 · C4–M1 · O2–M1 · C3–M2

Test IDs: SN009, SN001, SN004, SN022

Recording IDs are disjoint. SN001 test input was reconstructed from real EDF.

SN IDs are treated as subjects by the repository; person mapping is unavailable.

Current Model and Pipeline



Hybrid CNN–Transformer/self-attention–BiLSTM | 225,925 trainable parameters

Results Obtained — Raw vs Soft-Viterbi

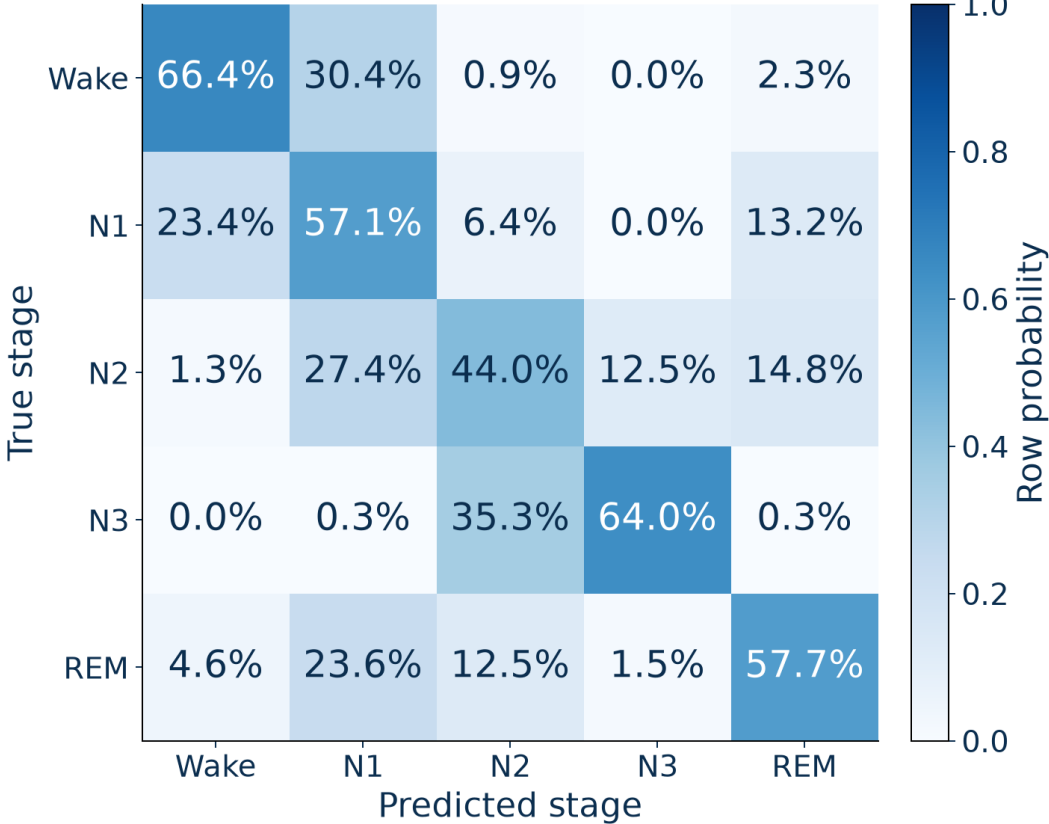
Metric	Raw model	Soft-Viterbi
Accuracy	53.75%	61.99%
Balanced accuracy	57.83%	64.26%
Macro F1	55.12%	61.76%
Weighted F1	54.92%	61.93%
Cohen's kappa	0.4060	0.4999

Same 3,773 test epochs from four held-out recording IDs.

Accuracy gain: +8.24 percentage points; every listed aggregate metric improved.

Results — Stage Errors and Uncertainty

Raw evidential model: row-normalized confusion matrix



Class order: Wake, N1, N2, N3, REM. HMC-PPT-20261007-01

Raw five-stage evaluation

Lowest F1: N1 = 0.366

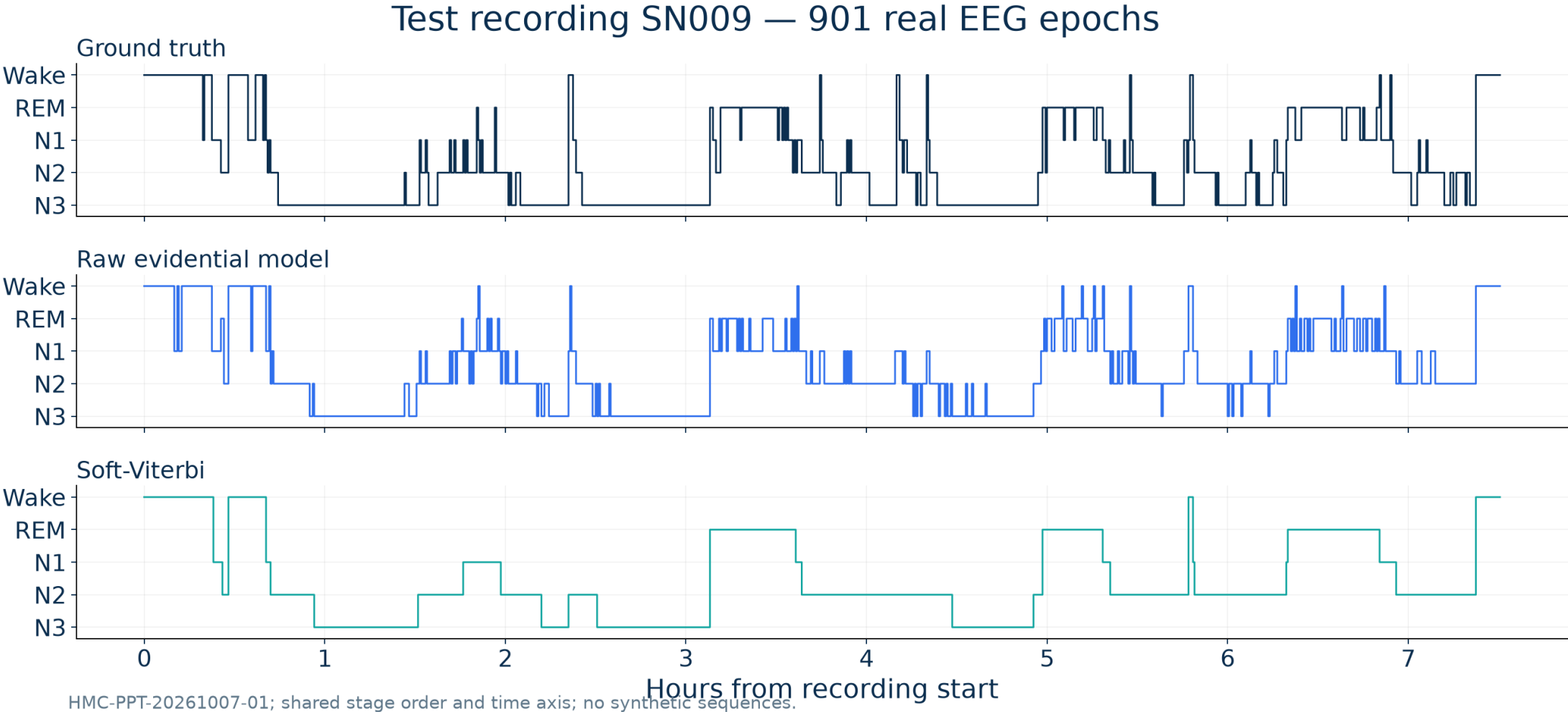
Lowest recall: N2 = 44.00%

Error-detection AUROC: 0.580

ECE (10 bins): 0.0705

Uncertainty separates errors weakly; reliable referral is future work.

Sequence Refinement — Real Test Recording SN009



Personalized Research Profiles

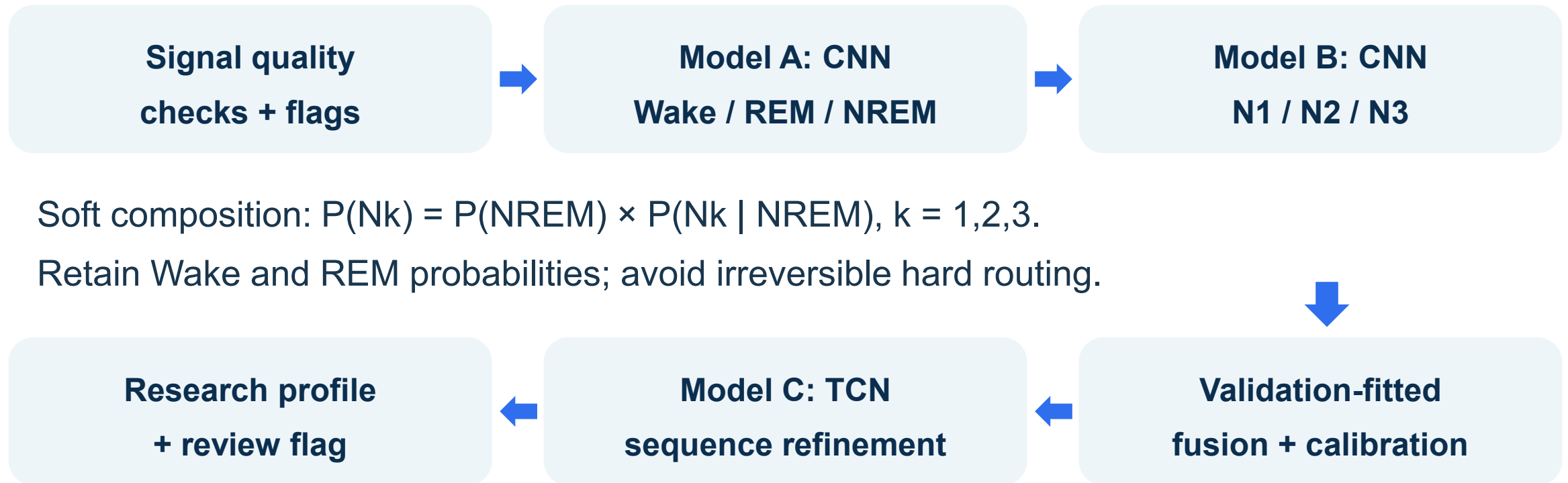
Recording	SE proxy	WASO (min)	N3 / all	SFI (/h)	Risk
SN009	90.01%	14.0	23.97%	1.18	Healthy
SN001	91.45%	31.5	24.71%	1.84	Healthy
SN004	88.48%	34.5	7.19%	2.27	Mild Risk
SN022	84.03%	74.0	9.08%	2.00	Mild Risk

Source: Soft-Viterbi labels only. Ground-truth profiles are stored separately.

Research category agreement with ground truth: 2/4 recordings.

SFI is a stage-transition proxy. Time in bed was not directly verified.

Future Work — Multistage, Multi-model Pipeline



Soft composition: $P(N_k) = P(\text{NREM}) \times P(N_k | \text{NREM})$, $k = 1, 2, 3$.

Retain Wake and REM probabilities; avoid irreversible hard routing.

Fuse hierarchical scores with the current five-stage model as a parallel baseline.

Add the sequence model only after measuring whether the experts help.

Evaluate stage recall, macro F1, calibration and cost; no future performance is claimed.

Conclusion and Future Priorities

What was demonstrated

One verified hybrid model and training-derived decoding: 53.75% raw → 61.99% decoded accuracy on 3,773 test epochs.

Next engineering milestone

Train coarse-stage and NREM experts; validate soft fusion against the current model. Then test a learned temporal refiner as an ablation.

What is required before stronger claims

Use grouped validation and a new locked test cohort; seek external validation and expert review of biomarker definitions and risk thresholds.

Current limits: four test recordings, historical test reuse, no clinical or external validation.